

Scenario Based Activity

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ITAD506 - Computer vision

1. A CCTV image contains salt-and-paper noise due to transmitter cross.
Median filter - Removes salt-and-paper noise effectively while preserving edges.
2. A satellite image is corrupted by Gaussian noise from sensor.
Gaussian filter - Reduces Gaussian noise by smoothing the image.
3. A medical x-ray image has random noise but fine edges must be preserved.
Bilateral filter - Removes random noise while preserving fine edges.
4. A scanned document has black and white speckles scattered randomly.
Median filter - Eliminates scattered black-and-white dots in scanned documents.
5. A low-light image shows heavy grain without clear boundaries.
Non-local means (NLM) filter - Reduces heavy grain / noise while preserving image details.

6. A foggy road image has very low contrast.
Histogram equalisation - Enhances the contrast of the foggy image.
7. An underwater image appears dull and washed out.
Contrast stretching - Improves brightness and contrast in dull underwater images.
8. A chest x-ray shows poor visibility of internal structure.
Clarke (Contrast limited adaptive histogram equalisation) - Enhances local contrast and improves visibility of internal chest x-ray.
9. A night-time traffic image lacks intensity variation.
Histogram equalisation - Increases intensity variation in night-time traffic image.
10. A gray scale image used only a narrow intensity range.
Contrast stretching - Expands the narrow gray intensity range to improve image visibility.